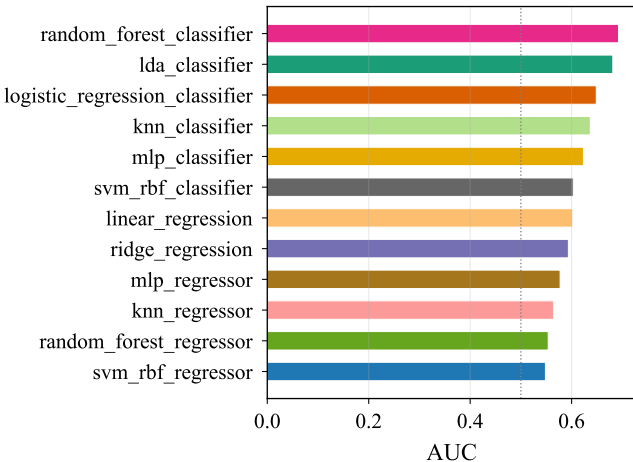


(a) Need-query classification



(b) Need-query F1

